3DCNN-NF: Few-Shot Hyperspectral Image Change Detection Based on 3-D Convolution Neural Network and Normalizing Flow

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***Abstract*— In recent years, deep learning has shown promising results in change detection (CD) of hyperspectral images (HSIs). However, these algorithms often require a large number of labeled samples, which can be costly in practical applications. This article proposes a few-shot HSI CD method based on a 3-D convolution neural network and normalizing flow. To mitigate the issue of limited training samples, we develop a low-parameter baseline CD model utilizing 3-D convolution neural networks to extract spectral–spatial features while preventing overfitting. In addition, the proposed method analyzes the spectral–spatial distribution of hyperspectral variation by normalizing flow and generates hyperspectral tensor samples that approximate the distribution, thus enhancing CD performance. Furthermore, to improve the sample generation speed, we employ soft labels and label smoothing techniques to assign high-quality labels to the generated samples, thereby increasing the number of available samples. The method is evaluated on three datasets, and the experimental results demonstrate the efficacy of the proposed approach for detecting changes with limited training samples. The proposed method offers a promising solution to the challenge of detecting changes with limited training samples in HSIs.**

***Index Terms*— 3-D convolution neural network, change detection (CD), few-shot, hyperspectral image (HSI), normalizing flow.**

I. Introduction

CHANGE detection (CD) based on remote sensing images is a crucial technology for comprehending Earth's transformations. This technology finds widespread applications in natural resource auditing, urban planning, disaster assessment, and various other scenarios [1], [2], [3], [4], [5]. The hyperspectral image (HSI), as a type of remote sensing imagery, has gained significance as a valuable data source for ground object observation due to its exceptionally high spectral resolution. Numerous methods have been proposed by researchers for CD using HSIs.

Previous studies introduced CD algorithms based on algebraic theory. HSIs contain abundant spectral information, making the spectral difference at different time points crucial for CD. Bovolo and Bruzzone [6] introduced the change vector analysis (CVA) method, which focuses on examining the spectral difference at different time points to determine the intensity and direction of spectral change, serving as the foundation for pixel CD. Although HSIs provide rich spectral information, their extremely high spectral resolution results in redundant spectral information. Consequently, feature extraction after reducing the dimension of hyperspectral data is a viable approach. Principal component analysis (PCA) is a widely employed method for dimensionality reduction and clustering. It involves calculating the eigenvalues and eigenvectors of the covariance matrix, extracting the eigenvectors corresponding to the largest eigenvalues, and employing them for data dimensionality reduction and clustering. Zhang et al. [7] applied PCA to extract features for dimensionality reduction and cluster analysis, ultimately assigning all tested pixels into change or nonchange classes. Another dimensionality reduction method is canonical correlation analysis (CCA), which finds representative vectors in low-dimensional space to reduce the dimensionality of vectors in two high-dimensional spaces [8]. Nielsen et al. [9], [10] presented two CD methods, namely, the multivariate CD (MAD) method and the CD method based on iterative weighting (IR-MAD). They calculated the weight and change of the MAD variable for pixel discrimination using CCA. The aforementioned methods are all rooted in algebraic and statistical theory, enabling unsupervised completion of the CD task and yielding favorable results.

However, traditional algebraic and statistical algorithms lack adaptability to high-dimensional feature spaces. Given the inherent high-dimensional properties of HSI, deep learning has emerged as an effective processing approach. For example, Gao et al. [11], [12], [13] introduced several deep

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learning-based approaches for remote sensing analysis, such as DFINet for coastal wetland classification, CSMFormer for image fusion and classification, and ACL-CNN for extracting complementary information from remote sensing data. In recent years, numerous studies have applied deep learning to HSI CD. Dong et al. [14] focused on the hyperspectral abundance matrix and proposed a hybrid attention network for subpixel-level correlation analysis. Huang et al. [15] integrated attention mechanisms into their model architecture to enhance feature extraction for CD. Zhao et al. [16] employed the recurrent neural network (RNN) for extracting spectral features from HSI. Moustafa et al. [17] applied techniques like random oversampling and bagging sets to tackle the problem and evaluated the versatility and performance of different UNet models. Hu et al. [18] introduced HyperNet, a pixel-level self-supervised network for understanding HSI spatial–spectral features. Zhao et al. [19] devised a CD method using a simplified 3D-CNN autoencoder, enabling unsupervised extraction of spectral–spatial features and reducing spectrum redundancy effectively. Roy et al. [20] combined 3D-CNN with 2D-CNN, allowing the joint learning of spectral–spatial features with 3D-CNN and abstract spatial representation with 2D-CNN built upon it. Song et al. [21] devised a novel cross-time interactive symmetric attention module for bi-temporal image interaction. They structured a Siamese network around this module, facilitating hierarchical extraction of change information symmetrically. Zhan et al. [22] integrated the Siamese network into the CD task, employing a deep neural network to extract spectral space feature vectors and redefining the CD problem as a similarity measurement problem between feature tensor pairs. Wang et al. [23] introduced a hybrid affinity matrix to better utilize subpixel-level end member abundance information, improving cross-channel feature extraction and providing a new dataset for evaluating CD algorithms. Luo et al. [24] presented DCENet, a semi-supervised approach for CD in multitemporal HSIs, utilizing a differential fusion attention subnetwork and dual-branch Siamese enhancement module to effectively leverage limited labeled samples and improve CD accuracy. Luo et al. [25] introduced MSDFFN, a novel multiscale diff-changed feature fusion network for HSI CD, enhancing feature representation by capturing refined change components between bitemporal HSIs across different scales, thereby achieving superior performance in detecting subtle changes compared to existing methods. Deep learning techniques can extract rich spectral and spatial features from HSIs, thereby reducing the impact of noise and redundancy on the CD algorithm to a certain extent and yielding promising results.

However, deep learning requires many known labeled samples for model learning, which is costly. In practical HSI CD, acquiring labeled samples is a high-cost process, typically necessitating manual annotation of a limited subset of pixels, commonly ranging from 3% to 20%. Although this percentage may not seem substantial, it still imposes a significant workload in real-world applications. Using a small sample size for labeling may escalate the risk of overfitting, potentially compromising the model's generalization performance. Therefore, we want to explore whether satisfactory CD results can be obtained under the condition of a small number of samples. Many parameters of the deep learning model bring the model a strong learning ability. If the model is directly trained with few samples, the model will inevitably fall into overfitting. Therefore, a model with small parameters can speed up training and reasoning and learn more generalized features. In addition, the small sample problem can also be solved by expanding the training samples, that is, data augmentation. Data augmentation essentially achieves the purpose of expanding the dataset by artificially introducing a prior about invariance. Commonly used data enhancement methods, including rotation, cropping, scaling, flipping, and adding noise, are widely used in image processing [26], [27]. There are also many related studies on the execution strategies of automatic search data augmentation algorithms [28], [29], [30]. Nevertheless, conventional data augmentation methods frequently utilized in image processing encounter constraints when addressing the intricate structures of high-dimensional tensors. Specific transformations introduced by certain traditional data augmentation methods, such as rotation or cropping, may occasionally disturb the intrinsic distributional characteristics of HSI data. Generative adversarial networks (GANs) are min–max games between a generator, which generates synthetic data given a random noise vector, and a discriminator, which distinguishes between actual data and the data generated by the generator [31], [32], [33]. GAN can implicitly learn the data distribution so that the generated samples are less destructive to the original data distribution. Since GAN does not explicitly give the distribution of data, and the training of GAN needs to achieve Nash equilibrium, this leads to insufficient training stability of GAN. Dinh et al. [34] proposed the normalizing flow called NICE, which provides a method to directly estimate the unknown overall data distribution from the known data distribution. The normalizing flow model can be updated using a conventional backpropagation algorithm, which makes its training easy. Recent researches provided several different normalizing flows and continued to improve the feature expression ability of the normalizing flows. Dinh et al. [35] utilized Real NVP transformations for unsupervised density estimation, ensuring exact log-likelihood computation, precise sampling, and interpretable latent spaces, demonstrated on four natural image datasets. Ho et al. [36] improved flow-based generative models by tackling limitations in dequantization, affine flows, and conditioning networks, achieving state-of-the-art nonautoregressive density estimation for images and narrowing the performance gap with autoregressive models. Kingma and Dhariwal [37] introduced glow, a generative flow model utilizing invertible 1×1 convolutions, which significantly improves log-likelihood and sample quality, showcasing efficient synthesis of large, realistic-looking images optimized for the plain log-likelihood objective. Huang et al. [38] introduced neural autoregressive flows, unifying normalizing flows and autoregressive models.

The overarching goal of this study is to devise a CD model exhibiting resilient generalization capabilities within the constraints of exceedingly scarce labeled samples. The predominant challenge encountered in pursuing this objective stems from the pronounced likelihood of neural networks

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succumbing to overfitting under these circumstances, thereby substantially compromising the model's generalization capacity. On one hand, we need to reduce the model's parameter count, and on the other hand, we need to augment samples while preserving the original sample distribution as much as possible. Consequently, accurately estimating the sample distribution and fostering synergy between the model and samples represent formidable challenges demanding attention. Therefore, we propose a few-shot HSI CD approach, incorporating a 3D-CNN and sample generation using normalizing flow. To address the challenge posed by a limited number of training samples, we develop a foundational CD model based on 3D-CNN with a low number of parameters. Leveraging the strong feature extraction capabilities of 3D-CNN for spectral–spatial features, we compress the model parameters and mitigate the risk of overfitting. Simultaneously, we employ normalizing flow technology to generate tensor samples that closely approximate identically distributed tensors for both the constant pixel tensor and the changed pixel tensor. We then filter the generated samples using the foundational CD model, incorporating them into the training set. Finally, we utilize this refined training set to train the foundational CD model, yielding accurate CD results.

The main contributions of this work are summarized as follows.

1) A simple 3D-CNN model for HSI CD is constructed using a small number of network parameters. Experiments demonstrate that the simple model maintains strong generalization performance despite the limited training data.
2) To better align the augmented samples with the original data distribution, we employ the normalizing flow to learn the distribution of the HSI change tensor and generate augmented samples accordingly. By integrating the original samples with the augmented ones, the model is trained to better capture the distribution of the HSI change tensor, thus enhancing the overall robustness of the model.
3) We employ a combination of soft labeling techniques with label smoothing to meticulously filter the generated samples and assign appropriate labels. This process ensures that the model is able to learn from high-quality data.

The structure of the article is organized as follows. In Section II, we provide an introduction to the fundamental concept of normalizing flow. Section III presents a detailed explanation of 3DCNN-NF. In Section IV, we present the datasets used, evaluation indices, experimental results, and provide an analysis of the experiments. Finally, in Section V, we conclude the article.

II. Background

A. Change of Variable Theorem

Let X and Z be random variables, and the corresponding probability density functions are $p(x)$ and $q(z)$, where $x, z \in \Re^n$. If there is a bijection $f{:}X \to Z$, then

$$q(z) = p(x)\left|\det \mathcal{J}_f^{-1}\right| \tag{1}$$

where $\mathcal{J}_f$ is the Jacobian matrix of f and $\det(\mathcal{J}_f) \neq 0$. This theorem provides the relationship between the probability density functions of two random variables and serves as the foundation for transforming a known probability distribution into an unknown one.

B. Principles of Normalizing Flow

The data generation model aims to learn the underlying distribution of the data from the extracted samples in order to generate high-quality data samples that follow the same distribution. Let $X \sim p_D(x)$ be the real data distribution we finally want to get, the samples obtained by this distribution are $\{x_1, x_2, \ldots, x_m\}$; the known distribution is $Z \sim \pi(z)$, bijection $G{:}Z \to X$, let $G(z) \sim p_G(x)$ be the generated data distribution. We hope that $p_D(x)$ and $p_G(x)$ are as close as possible, that is, the KL divergence between $p_D(x)$ and $p_G(x)$ is as small as possible

$$G^* = \arg\min_G D_{\mathrm{KL}}(p_D(x)||p_G(x)). \tag{2}$$

Minimizing the KL divergence is approximately equivalent to maximizing the log-likelihood function, so the optimization problem is transformed into

$$G^* \approx \arg\max_G \sum_{i=1}^{m} \log p_G(x_i). \tag{3}$$

According to the change of variable theorem, we can get

$$G^* \approx \arg\max_G \sum_{i=1}^{m} \log \pi\left(G^{-1}(x_i)\right) + \log\left|\det \mathcal{J}_G^{-1}\right|. \tag{4}$$

It can be seen from (4) that the bijective G needs to be easily inverted. The Jacobian determinant of its inverse transformation also needs to be easily obtained, which significantly limits the feature expression ability of G. Therefore, it is possible to gradually transform from a known simple distribution to a complex distribution by superimposing multiple bijective transformations to improve the feature expression ability.

III. Proposed Method

A. Base CD Model

Let $X^{(1)} \in \Re^{h\times w\times l}$ and $X^{(2)} \in \Re^{h\times w\times l}$ represent HSIs acquired at two times. We use the absolute value difference to obtain the change range map X

$$X = \left|X^{(2)} - X^{(1)}\right|. \tag{5}$$

For the ith pixel ($i = 1, 2, \ldots, hw$), extract the spectrum of neighboring pixels centered on this pixel and combine them into a tensor $X_i \in \Re^{b\times b\times l}$. The tensor combines the pixel's neighborhood spectral and spatial information, weakening the noise's influence on the spectrum. X_i is a 3-D hyperspectral data tensor, and we use 3D-CNN to extract its spectral and spatial fusion features. In 3D-CNN, the value v_{mn}^{xyz} of the position (x, y, z) in the nth feature map of the mth layer is

$$v_{mn}^{xyz} = \varphi\left(\sum_k \sum_{p=0}^{P_m-1} \sum_{q=0}^{Q_m-1} \sum_{r=0}^{R_m-1} w_{mnk}^{pqr} \times v_{(m-1)k}^{(x+p)(y+q)(z+r)} + b_{mn}\right) \tag{6}$$

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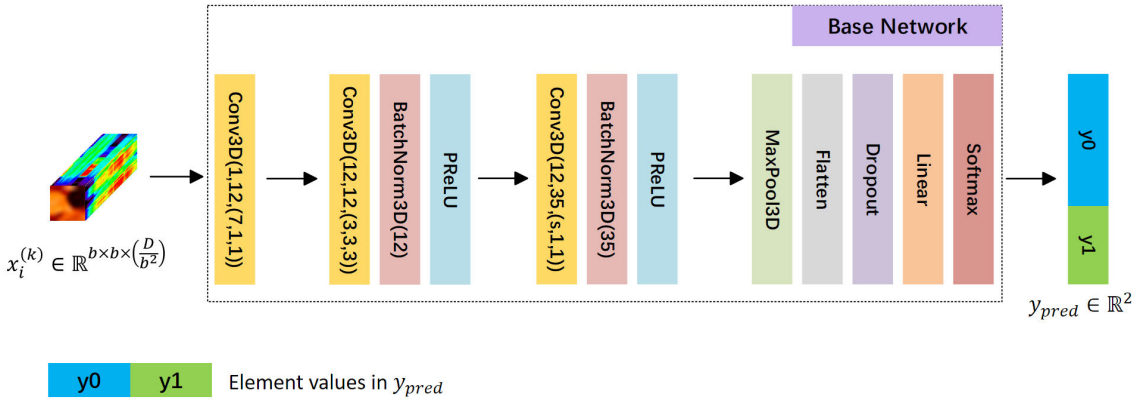


Fig. 1. Base network for CD.

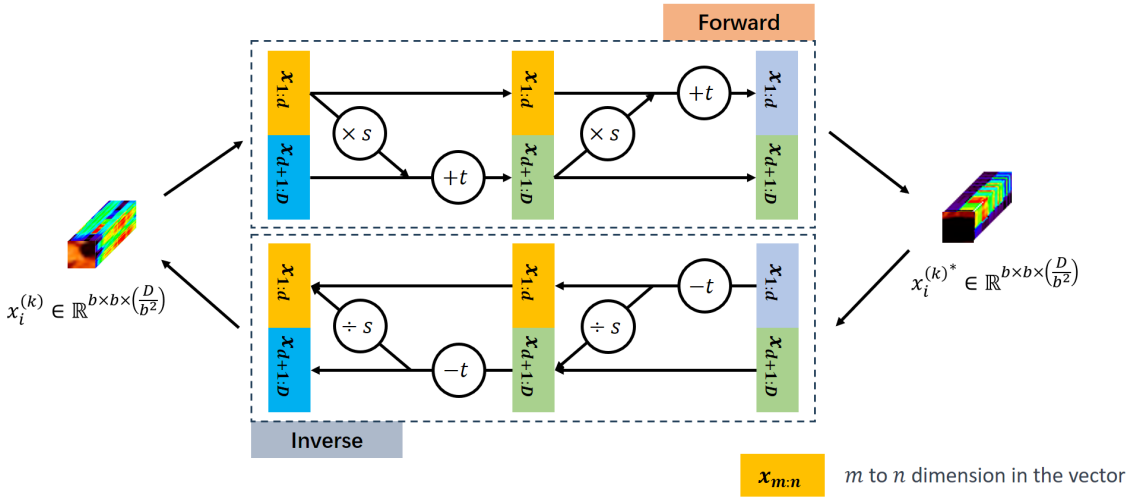


Fig. 2. Flow-based generation model.

where $\varphi(\cdot)$ is the activation function, b_{mn} is the bias, P_m, Q_m, and R_m correspond to the width, height, and depth of the convolution kernel, respectively, and w_{mnk}^{pqr} is the weight. 3D-CNN has demonstrated powerful feature learning capabilities in many HSI analysis tasks. However, in the case of a few training samples, models with complex parameters are prone to overfitting. Therefore, the basic CD model shown in Fig. 1 was adopted in this study. This model aims to use a CNN with few parameters to learn changing spatial–spectral features and prevent the model from overfitting as much as possible. Let the basic CD model be F_{base}, then

$$y_{\text{pred}} = \left[y_0, y_1\right] = F_{\text{base}}(X_i) \tag{7}$$

where y_{pred} represents the probability that the tensor X_i belongs to each category. Category 0 represents the unchanged category, and category 1 represents the variable category. So the category c_i to which X_i belongs is

$$c_i = \arg\max_i y_{\text{pred}}. \tag{8}$$

B. Generate Training Samples

Due to the small number of training samples with labels, the model may only partially describe each category, and it is elementary to fall into overfitting. Therefore, we plan to generate samples for each category to improve the model's ability to describe each category.

Fig. 2 shows the flow-based generation method. Let $X_i^{(c)} \in \Re^{b\times b\times l}$ be the training samples labeled c. We use the normalizing flow to model all $X_i^{(c)}$, based on reversible transformation, the known basic distribution is transformed into the distribution of category c. Then, a generated sample of category c can be obtained by sampling from the known basic distribution and performing the reversible transformation. Considering the speed and quality of generation, we adopted the affine coupling transformation in real-NVP as the reversible transformation in this study. First, transform $X_i^{(c)}$ into $x_i^{(c)} \in \Re^D$

$$x_i^{(c)} = \text{Reshape}\left(X_i^{(c)}, D\right) \tag{9}$$

where

$$D = \begin{cases} \text{bbl}, & l = 2k,\ k \in N^* \\ \text{bb}(l-1), & l = 2k-1,\ k \in N^*. \end{cases} \tag{10}$$

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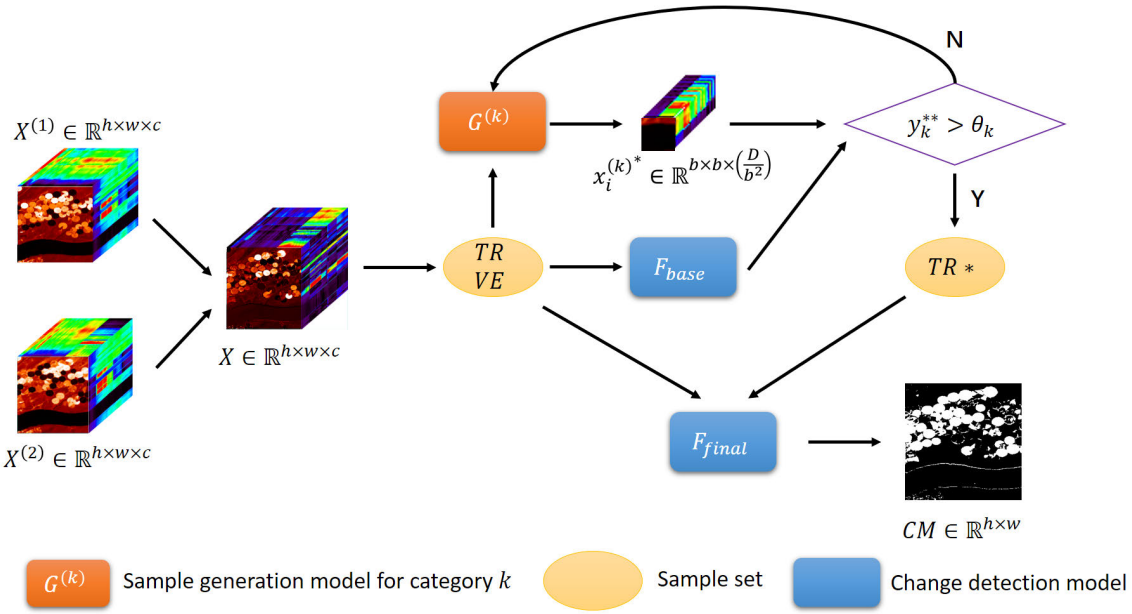


Fig. 3. 3DCNN-NF algorithm.

Performing affine coupling transformation on $x_i^{(c)}$

$$\begin{cases} \left(x_i^{(c)}\right)_{1:d} \leftarrow \left(x_i^{(c)}\right)_{1:d} \\ \left(x_i^{(c)}\right)_{d+1:D} \leftarrow \left(x_i^{(c)}\right)_{d+1:D} \odot s_1\left(\left(x_i^{(c)}\right)_{1:d}\right) \\ \qquad\qquad + t_1\left(\left(x_i^{(c)}\right)_{1:d}\right) \end{cases} \tag{11}$$

where $(x_i^{(c)})_{1:d}$ represents the dimension 1 to d of $x_i^{(c)}$, $d = (D/2)$, $\odot$ represents the Hadamard product, and both $s_1(\cdot)$ and $t_1(\cdot)$ are MLP functions. The MLP function can be expressed as

$$\mathrm{MLP}(x) = \tanh(W_3 \cdot \tanh(W_2 \cdot \tanh(W_1 \cdot x + b_1) + b_2) + b_3) \tag{12}$$

where W_i and b_i represent weight and bias, $i \in \{1, 2, 3\}$. The inverse of this transformation is

$$\begin{cases} \left(x_i^{(c)}\right)_{1:d} \leftarrow \left(x_i^{(c)}\right)_{1:d} \\ \left(x_i^{(c)}\right)_{d+1:D} \leftarrow \left(\left(x_i^{(c)}\right)_{d+1:D} - t_1\left(\left(x_i^{(c)}\right)_{1:d}\right)\right) \\ \qquad\qquad \oslash s_1\left(\left(x_i^{(c)}\right)_{1:d}\right) \end{cases} \tag{13}$$

where $\oslash$ denotes element-wise division. The Jacobian matrix $\mathcal{J}$ of this transformation is also easy to compute

$$\mathcal{J} = \begin{bmatrix} I_d & 0 \\ \dfrac{\partial x_{d+1:D}}{\partial x_{1:d}^T} & \mathrm{diag}(s_1(x_{1:d})) \end{bmatrix} \tag{14}$$

$$\det \mathcal{J} = \prod s_1(x_{1:d}). \tag{15}$$

In order to achieve a more adequate feature expression, an affine coupling transformation can also be performed on $x_i^{(c)}$

$$\begin{cases} \left(x_i^{(c)}\right)_{d+1:D} \leftarrow \left(x_i^{(c)}\right)_{d+1:D} \\ \left(x_i^{(c)}\right)_{1:d} \leftarrow \left(x_i^{(c)}\right)_{1:d} \odot s_1\left(\left(x_i^{(c)}\right)_{d+1:D}\right) \\ \qquad\qquad + t_1\left(\left(x_i^{(c)}\right)_{d+1:D}\right). \end{cases} \tag{16}$$

Similar to the derivation of (13)–(15), the transformation (16) is still invertible and the Jacobian matrix is easy to find. By alternately transforming (11) and (16), the feature coupling ability of the generative model can be significantly improved.

In this article, the standard logistic distribution $L(\cdot)$ is used as the known basic distribution. The normalizing flow generation model $G(\cdot)$ is obtained by solving the maximum likelihood optimization problem with the backpropagation algorithm. The generation loss function is

$$\text{generate loss} = \sum_{i=1}^{m} \log\left(L\left(G^{-1}\left(x_i^{(c)}\right)\right)\right) + \log\left|\det \mathcal{J}_G^{-1}\right|. \tag{17}$$

After training the generative model G, sample z_i from the standard logistic distribution and input it into the generative model to get the generated sample $x_i^{(c)^*}$, and then restore the tensor form $X_i^{(c)^*}$

$$X_i^{(c)^*} = \mathrm{Reshape}\left(G(z_i), \left(b, b, \frac{D}{b^2}\right)\right) \tag{18}$$

$X_i^{(c)^*}$ may not be used as a high-quality generated sample, and it is necessary to filter $X_i^{(c)^*}$ through F_{base}. Input $X_i^{(c)^*}$ into F_{base} to obtain the probability y_c^* of basic CD to identify the sample as category c

$$y_c^* = F_{\text{base}}\left(X_i^{(c)^*}\right). \tag{19}$$

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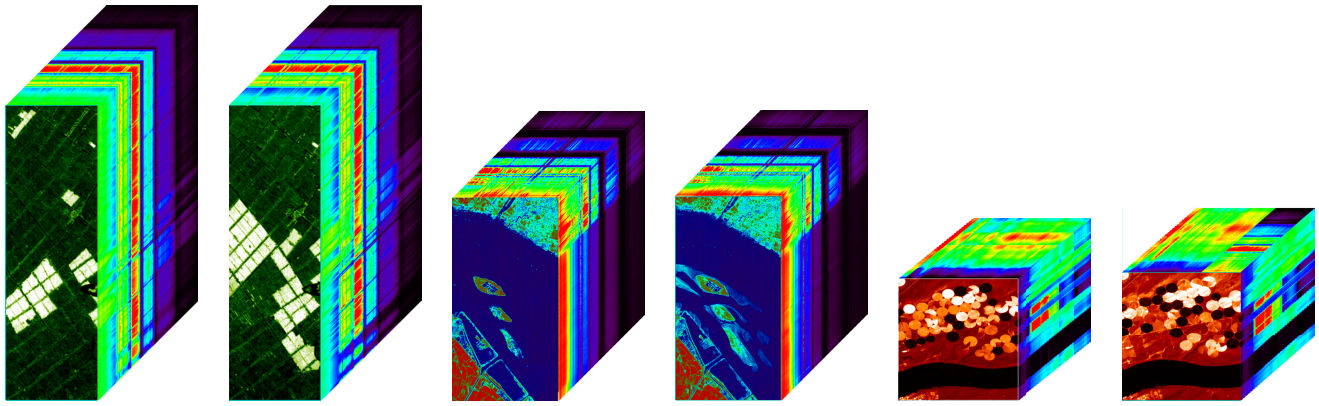


Fig. 4. Experiment Datasets. (a) Farmland on May 3, 2006. (b) Farmland on April 23, 2007. (c) River on May 3, 2013. (d) River on December 31, 2013. (e) USA on May 1, 2004. (f) USA on May 8, 2007.

An extended sample is selected only when the value of y_c^* exceeds a given threshold θ, and its label is assigned as the sample class c. Increasing the value of θ enables the CD model to identify more strong features in the generated samples. However, a larger value of θ increases the likelihood of discarding the generated samples, hindering the generation of a large number of samples. To address this issue, it is necessary to appropriately relax the acceptance criteria for the generated samples. Primarily, samples generated by the c-class generative model must be identifiable as class c, implying that θ must not be less than 0.5. Moreover, evaluating the strength of the features corresponding to the current category in the samples is necessary to enhance the model's learning of category features. Consequently, category hard labels are inadequate, and soft labels are employed instead. For each generated sample $X_i^{(c)^*}$ received, the label y^* is directly assigned to the generated sample. Soft labels enable the characterization of the strength of the category features present in the generated samples. As the configuration of soft labels relies on F_{base}, which only learns the features from a small subset of the dataset, the soft labels may exhibit some bias. To account for this bias in the soft labels, a label smoothing strategy is employed [39], [40], [41], [42]. The soft label y^* undergoes label smoothing to obtain the label y^{**}, which is given by

$$y_c^{**} = (1-\epsilon)y_c^* + \frac{\epsilon}{k} \tag{20}$$

where ϵ is the label smoothing hyperparameter.

Fig. 3 and Algorithm 1 demonstrate our proposed algorithm.

Algorithm 1 3DCNN-NF Algorithm

Require: HSIs $X^{(1)}$ and $X^{(2)}$, training sample set TR, verification sample set VE, kth class sample generation demand Q_k, threshold θ_k, label smoothing hyperparameter ϵ.

Ensure: CD result map CM

```
Calculate the difference map X = |X^(2) − X^(1)|.
Use TR to train the basic CD model F_base, and use VE to screen out the optimal model.
for k = 1, 2 do
    Transform all X_i^(k) in TR into x_i^(k).
    Use TR to train the generative model G^(k)(·) of category k, and select the optimal model through generate loss.
    Let the number of generated samples Q ← 0.
    while Q < Q_k do
        Randomly sample z_i from the standard logistic distribution L(·).
        Generate the sample X_i^(k)* and calculate y_k^* = F_base(X_i^(k)*).
        Calculate y_k^** = (1 − ε)y_k^* + ε/k
        if y_k^** ≥ θ_k then
            Add this sample to the generated training sample TR*
            Q ← Q + 1.
        end if
    end while
end for
Use TR and TR* to train the final CD model F_final, and use VE to screen out the optimal model.
Extract the tensor X_i corresponding to each pixel from X, input it to F_final to obtain all pixel CD results and generate a CD result map CM ∈ ℜ^{h×w}
```

IV. Experimental Results and Discussion

To assess the effectiveness of the proposed algorithm in the CD task, this chapter presents a comprehensive evaluation. First, we introduce three sets of real HSI datasets utilized in the experiments. Next, we outline the evaluation metrics employed to assess the performance of various CD algorithms. Finally, we present the experimental results and provide an in-depth analysis of each dataset.

A. Experiment Datasets

We utilized three real HSI datasets obtained from the Earth Observing-1 (EO-1) Hyperion sensor for our experiments. The datasets are Farmland [43], River [23], and USA [44], as depicted in Fig. 4. The Farmland dataset was taken from farmland near Yancheng City, Jiangsu Province, China, with prechange images dated May 3, 2006 and postchange images dated April 23, 2007. The River dataset was taken from a river and the area along it in Jiangsu Province, where the prechange image was dated May 3, 2013 and the postchange image was

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TABLE I
DETAILS OF THE DATASET

dataset	height	width	bands
Farmland	450	140	155
River	463	241	198
USA	307	241	154

TABLE II
CONFUSION MATRIX

confusion matrix		predict	
		positive	negative
real	positive	TP	FN
	negative	FP	TN

dated December 31, 2013. The USA dataset was taken from farmland in the city of Elmiston, Umatilla County, OR, USA, where the prechange image was dated May 1, 2004 and the postchange image was dated May 8, 2007. The EO-1 sensor captures electromagnetic waves in the wavelength range of 0.4–2.5 μm, with a spectral resolution of 10 nm. The spatial resolution of the sensor is approximately 30 m, and it consists of 242 spectral bands. These datasets encompass diverse land cover types, including farmland and rivers, enabling comprehensive evaluation of the CD algorithm.

The specific parameters of the public datasets used in this article are presented in Table I.

B. Evaluation Index

In this study, we employed several key indicators to evaluate the performance of CD algorithms. These indicators include the overall accuracy, Kappa coefficient, and F_1 score, which were calculated from the confusion matrix of the CD task.

The confusion matrix for the binary CD problem is presented in Table II.

The first evaluation metric adopted is OA, which evaluates the proportion of pixels accurately classified by the CD model

$$\mathrm{OA} = \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{TN}}. \tag{21}$$

The second evaluation metric employed is the kappa coefficient, which assesses the consistency of the CD model in handling each category. This metric holds particular significance in classification problems characterized by imbalanced sample distributions

$$p = \frac{(\mathrm{TP}+\mathrm{FP})(\mathrm{TP}+\mathrm{FN})+(\mathrm{TN}+\mathrm{FP})(\mathrm{TN}+\mathrm{FN})}{(\mathrm{TP}+\mathrm{TN}+\mathrm{FP}+\mathrm{FN})^2} \tag{22}$$

$$\mathrm{Kappa} = \frac{\mathrm{OA}-p}{1-p}. \tag{23}$$

The third evaluation metric utilized is the F_1 score, which considers both the precision and recall rates of the model simultaneously. It serves as a comprehensive measure for assessing the performance of the CD model

$$\mathrm{precision} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \tag{24}$$

$$\mathrm{recall} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \tag{25}$$

$$F_1 = \frac{2\times\mathrm{precision}\times\mathrm{recall}}{\mathrm{precision}+\mathrm{recall}}. \tag{26}$$

TABLE III
CD ALGORITHM COMPARISON EXPERIMENT RESULTS

Algorithm	Index	Dataset		
		Farmland	River	USA
SVM	OA	0.9713	0.9521	0.94
	Kappa	0.9307	0.6903	0.8175
	F1	0.9509	0.7162	0.8547
GETNET	OA	0.9415	0.939	0.8612
	Kappa	0.9095	0.6315	0.6267
	F1	0.8894	0.6744	0.7137
Hybrid-SN	OA	0.9667	0.9206	0.925
	Kappa	0.9181	0.6585	0.7668
	F1	0.9413	0.6017	0.8126
TDSSC	OA	0.9456	0.9414	0.8898
	Kappa	0.8592	0.7055	0.6195
	F1	0.8949	0.7363	0.6779
BRCN-ERN	OA	0.9671	0.9197	0.9452
	Kappa	0.9216	0.6099	0.8422
	F1	0.9451	0.6521	0.8775
CSANet	OA	0.9364	0.9415	0.9146
	Kappa	0.8502	0.6279	0.7418
	F1	0.8958	0.6599	0.7953
BiG-FSLF	OA	0.9496	0.9175	0.9122
	Kappa	0.8797	0.5759	0.7615
	F1	0.9076	0.6111	0.8156
FSCF-SSL	OA	0.9732	0.9606	0.9532
	Kappa	0.9358	0.7793	0.8626
	F1	0.9548	0.7675	0.8922
3DCNN-NF (base)	OA	0.9730	0.9601	0.9503
	Kappa	0.9355	0.7817	0.8570
	F1	0.9542	0.7750	0.8894
3DCNN-NF (final)	OA	**0.9740**	**0.9667**	**0.9539**
	Kappa	**0.9402**	**0.7986**	**0.8661**
	F1	**0.9560**	**0.8060**	**0.8950**

C. Experiment Settings

To comprehensively validate the effectiveness of the algorithm, the experiments in this article are divided into two parts.

In the first part, we initially compare our proposed basic CD model with existing supervised learning-based CD algorithms under few-shot conditions. This comparison aims to demonstrate the effectiveness of our proposed basic CD model.

In the second part, we utilize our proposed basic CD model as a benchmark and compare the sample generation method proposed in this article with other methods suitable for high-dimensional tensors. The objective is to illustrate the effectiveness of our proposed sample generation method.

Furthermore, we conduct parametric experiments on essential parameters in the proposed algorithm to analyze their sensitivity.

The proposed algorithm was implemented using Python 3.6 and PyTorch 1.8.0. The coding was conducted on a server equipped with a GeForce RTX 3090 GPU. Experimental findings corroborate the notion that in few-shot scenarios, diverse training samples exert a notable influence on the model's

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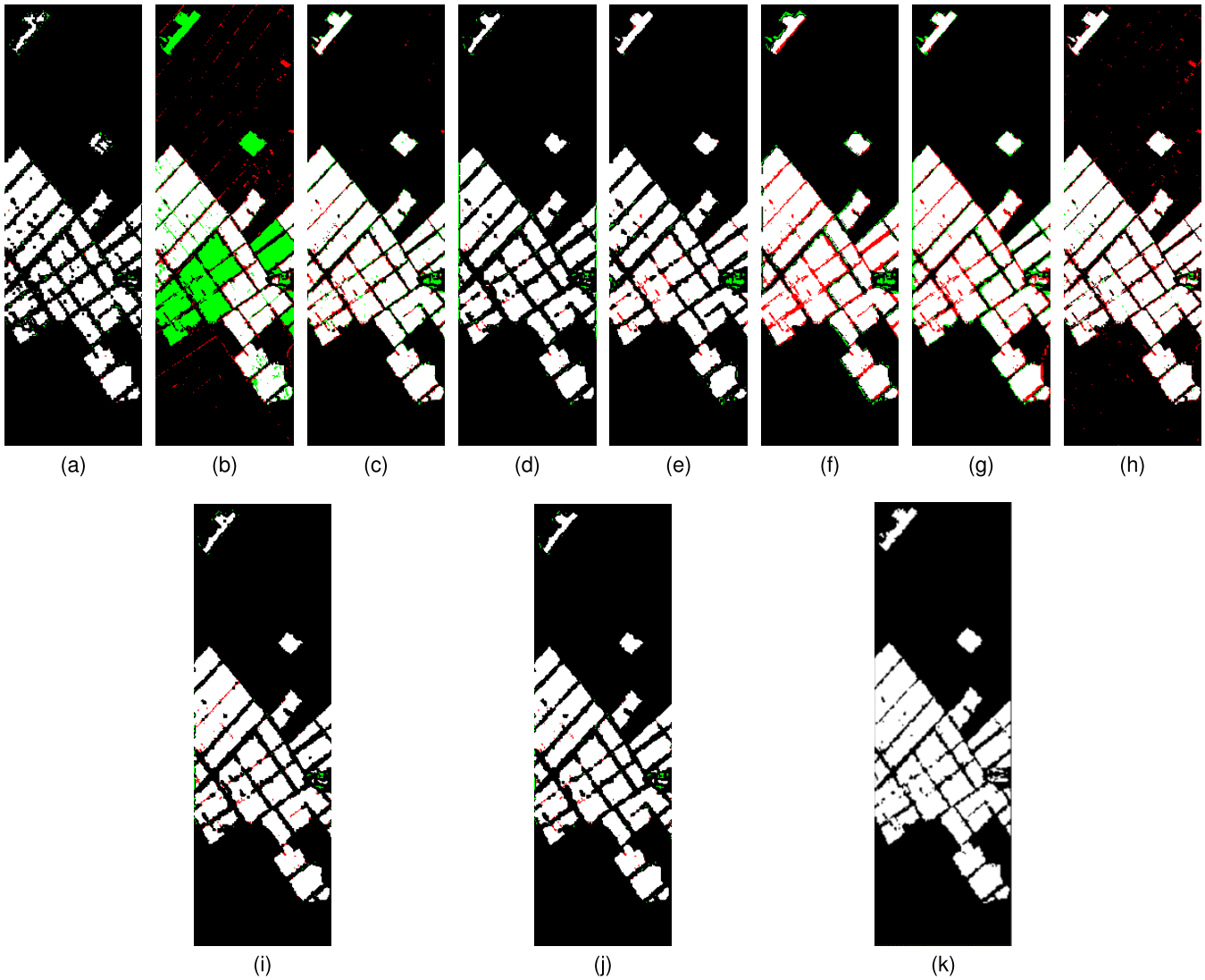


Fig. 5. CD algorithm comparison experiment result graph on farmland (TP: white; TN: black; FP: red; and FN: green). (a) SVM. (b) GETNET. (c) Hybrid-SN. (d) TDSSC. (e) BRCN-ERN. (f) CSANet. (g) BiG-FSLF. (h) FSCF-SSL. (i) 3DCNN-NF (base). (j) 3DCNN-NF (final). (k) Ground truth.

performance. The selection of high-quality initial samples is of paramount importance for both the initial model training phase and subsequent data augmentation procedures. For the three datasets, we compute the pure pixel index (PPI) for each pixel in each dataset [45]. We then select 40 unchanged pixels with high PPI values and 20 changed pixels with high PPI values as labeled training samples. In addition, we randomly select 30 samples for validation. To ensure a fair comparison as much as possible, for the comparative algorithms that do not use a validation set, we incorporate the validation set into the training set during model training. The batch size for the basic CD model was set to 32. Considering the empirical knowledge from existing patch-based HSI CD methods and limitations in computational resources, we specify $b = 3$ in the experiments.

The Adam optimizer with a learning rate of 0.001 and weight decay of 0.9 was utilized. Early stopping technology was employed, with the F_1 score serving as the monitoring target. The training was terminated if there was no improvement in the monitoring target over 100 epochs.

Regarding the generated model, a batch size of 30 was used, with Adam as the optimizer, a learning rate of 0.0002, and a weight decay of 0.9. In early stopping, the generation loss was used as the monitoring target, and training was stopped if there was no decrease in the generation loss over 5000 epochs.

D. Basic CD Model Experiments

To assess the progress and effectiveness of the basic CD model in the context of few training samples, we conducted a comparison with several advanced supervised CD algorithms, namely, SVM [46], Hybrid-SN [20], GETNET [23], TDSSC [47], BRCN-ERN [48], CSANet [21], BiG-FSLF [49], and FSCF-SSL [50]. The evaluation index values and result graphs of this comparison, along with those of the proposed algorithm, are presented in Table III and Figs. 5–7. The test time and network parameters of all algorithms are shown in Table IV.

The Farmland dataset primarily exhibits changes in agricultural areas, which are often concentrated. Therefore, CD algorithms tend to perform better when there are few

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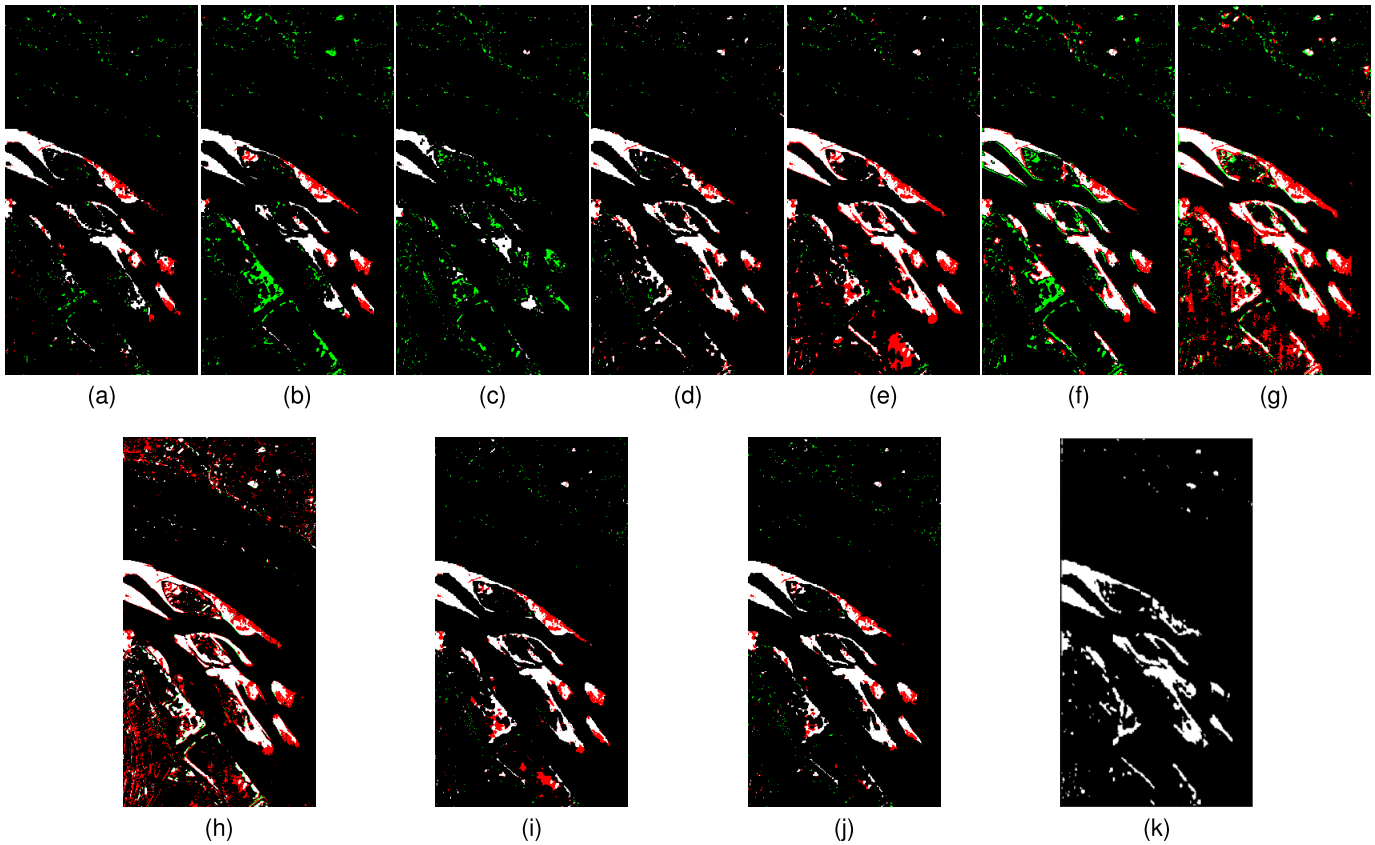


Fig. 6. CD algorithm comparison experiment result graph on the river (TP: white; TN: black; FP: red; and FN: green). (a) SVM. (b) GETNET. (c) Hybrid-SN. (d) TDSSC. (e) BRCN-ERN. (f) CSANet. (g) BiG-FSLF. (h) FSCF-SSL. (i) 3DCNN-NF (base). (j) 3DCNN-NF (final). (k) Ground truth.

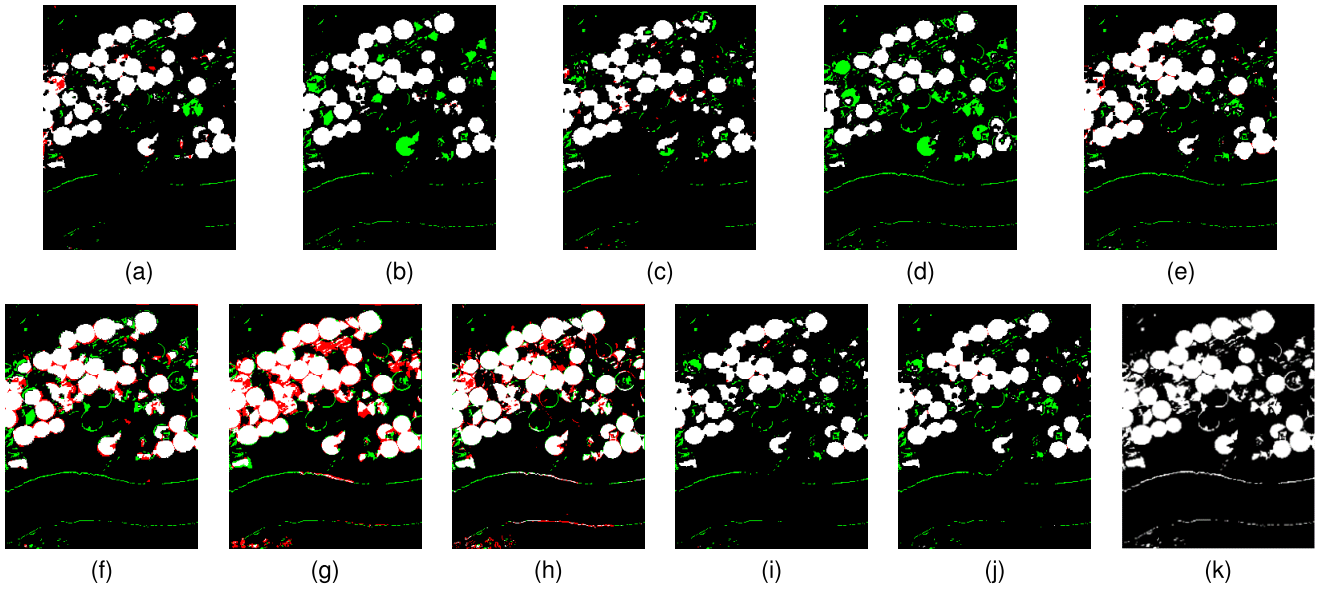


Fig. 7. CD algorithm comparison experiment result graph on USA (TP: white; TN: black; FP: red; and FN: green). (a) SVM. (b) GETNET. (c) Hybrid-SN. (d) TDSSC. (e) BRCN-ERN. (f) CSANet. (g) BiG-FSLF. (h) FSCF-SSL. (i) 3DCNN-NF (base). (j) 3DCNN-NF (final). (k) Ground truth.

training samples. The CD results graph reveals that the effectiveness of CD is influenced by small irregular change areas and false alarms caused by noise interference. The performance of CD is influenced by two issues: false alarms along the boundaries of farmland and missed detections of variations within the water flow regions, which are used for comparative algorithms. Our proposed algorithm demonstrates superior overall performance in addressing both of these issues, and further suppresses false alarms along boundaries through the utilization of sample augmentation techniques.

The River dataset captures regional and material changes along rivers and their banks. It has the widest spatial range and the largest number of bands among the three experimental datasets, which translates to a lower proportion of training

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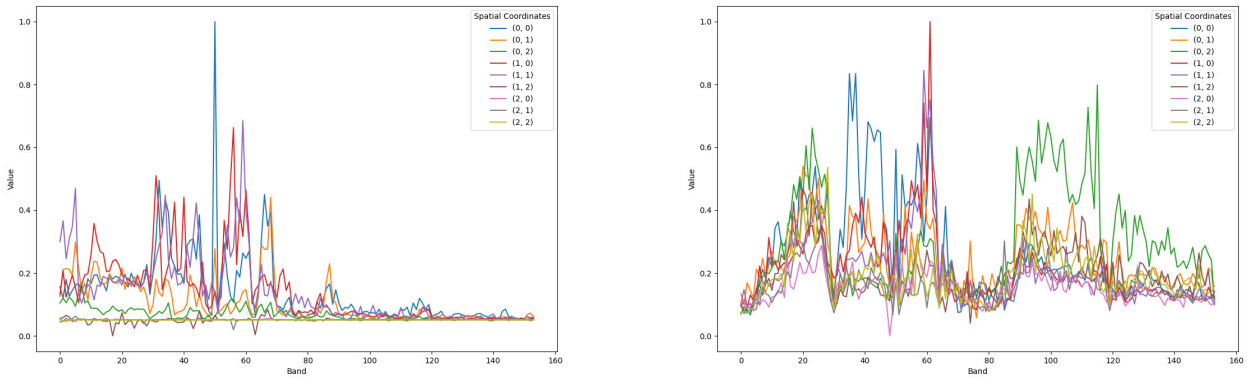


Fig. 8. Sample generation example in Farmland Dataset. (a) Label: [0.9300, 0.0699]. (b) Label: [0.0500, 0.9500].

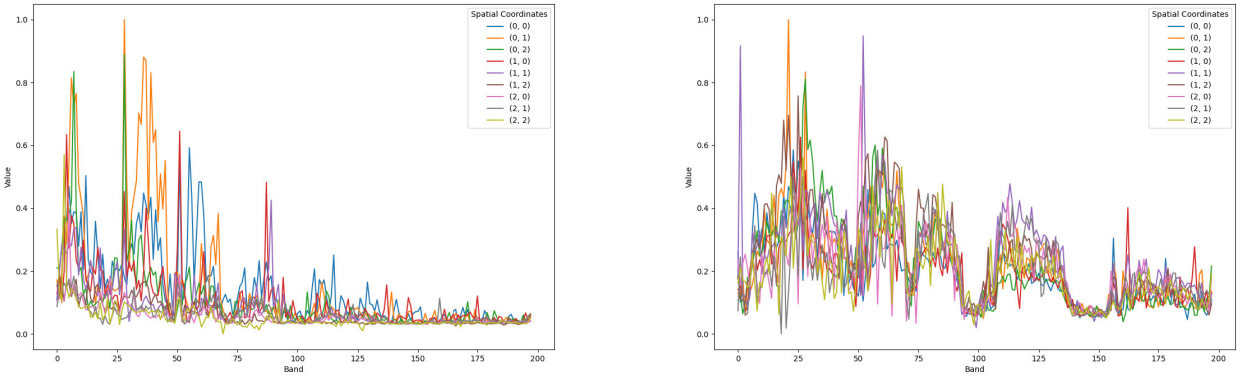


Fig. 9. Sample generation example in River Dataset. (a) Label: [0.9479, 0.0520]. (b) Label: [0.0515, 0.9485].

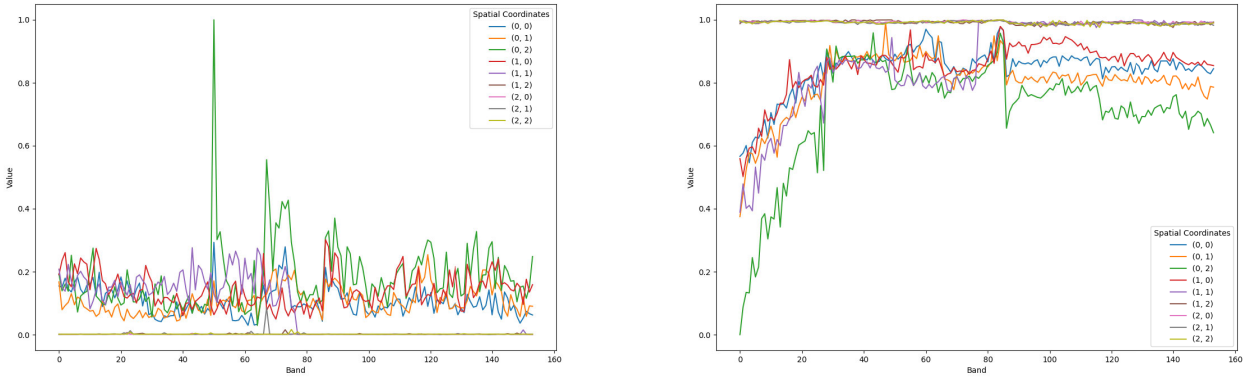


Fig. 10. Sample generation example in USA Dataset. (a) Label: [0.9259, 0.0740]. (b) Label: [0.0500, 0.9500].

samples. The ground truth map indicates that the changing area in the River dataset is irregular and scattered, posing additional challenges for CD. The CD result graph reveals significant differentiation in the detection results of various CD algorithms. From the detection results, it is evident that the comparative algorithm exhibits a significant number of false alarms and missed detections when handling intricate and scattered material changes, indicating inadequate generalization performance of the model. In contrast, our proposed algorithm significantly suppresses false alarms and missed detections, demonstrating the robustness of our proposed approach.

The USA dataset encompasses changes in irrigated farmland and rivers, offering a comprehensive test case. The main challenge lies in detecting changes in river edge areas and subtle changes within farmland areas. From the CD maps, it is

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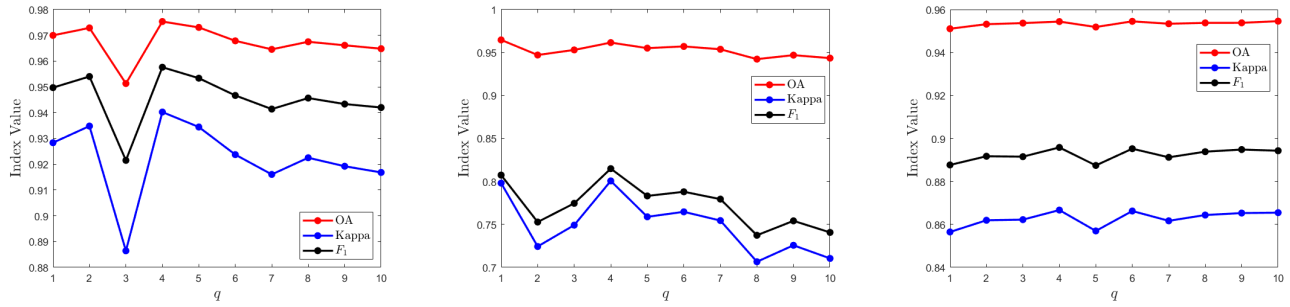


Fig. 11. Experimental results of Q_k value on three datasets. (a) Farmland. (b) River. (c) USA.

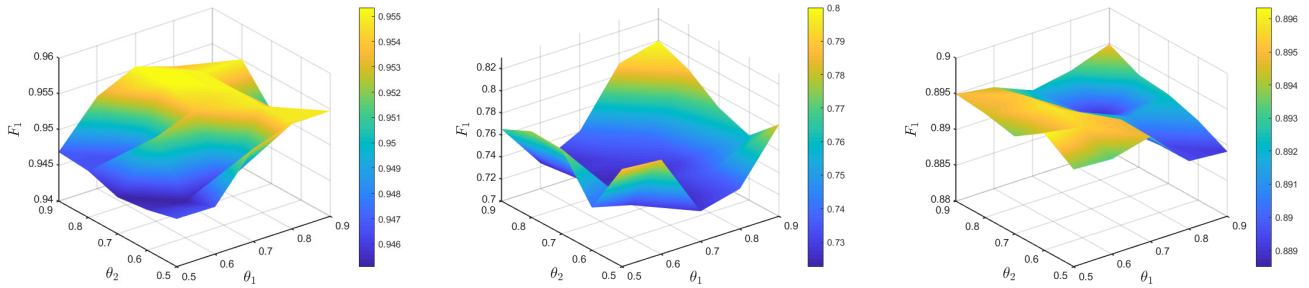


Fig. 12. Hard label results of θ_k value on three datasets. (a) Farmland. (b) River. (c) USA.

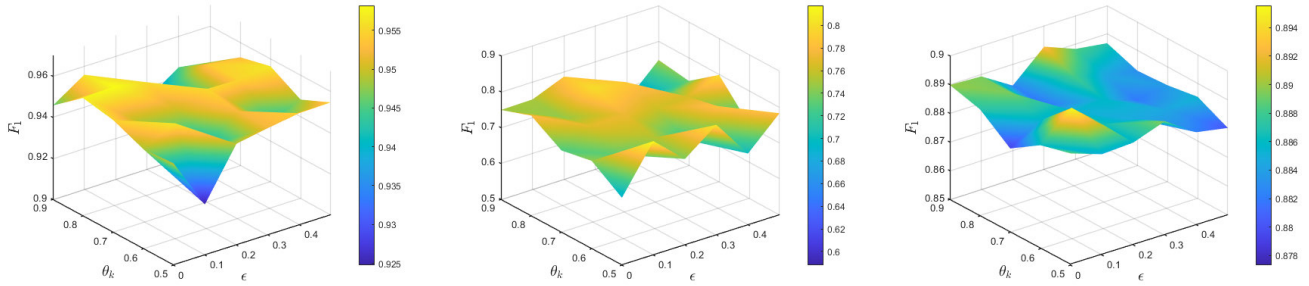


Fig. 13. F_1 results of θ_k and ϵ value on three datasets. (a) Farmland. (b) River. (c) USA.

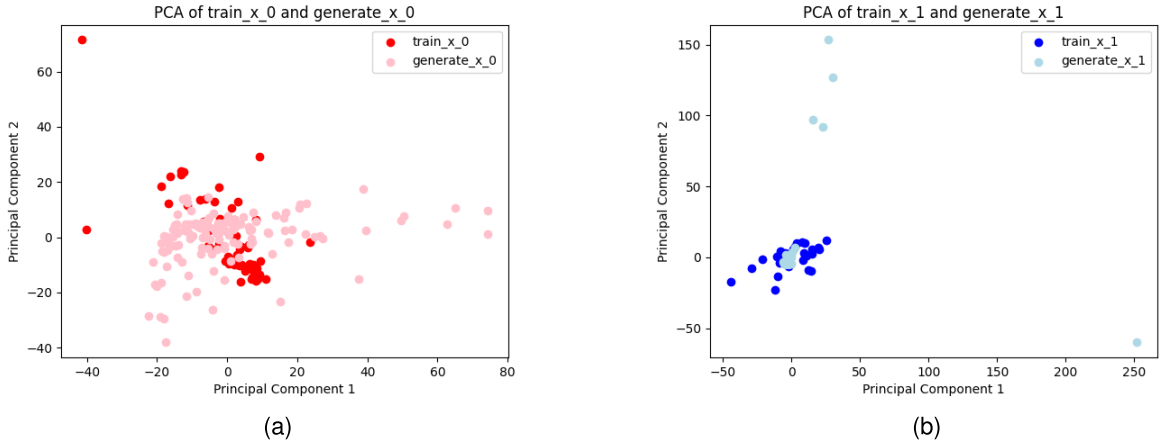


Fig. 14. Scatter plot of sample distribution for the Farmland dataset (train_x_0 represents real unchanged samples, generate_x_0 represents generated unchanged samples; train_x_1 represents real changed samples, generate_x_1 represents generated changed samples). (a) Unchanged samples. (b) Changed samples.

evident that all algorithms on this dataset exhibit a significant number of missed detections, particularly concerning changes along river boundaries. However, our proposed algorithm achieves a significantly lower missed detection rate while also substantially reducing false alarms, resulting in a substantial improvement in the effectiveness of CD.

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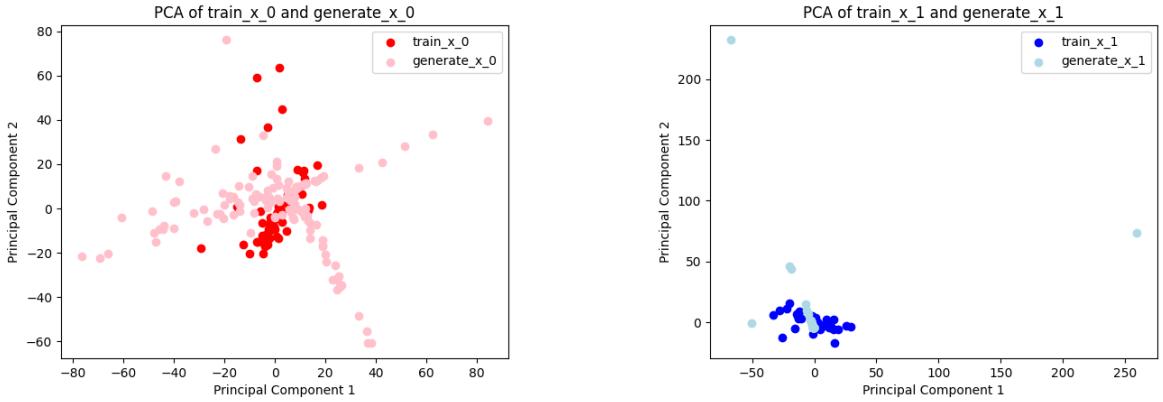


Fig. 15. Scatter plot of sample distribution for the River dataset (train_x_0 represents real unchanged samples, generate_x_0 represents generated unchanged samples; train_x_1 represents real changed samples, generate_x_1 represents generated changed samples). (a) Unchanged samples. (b) Changed samples.

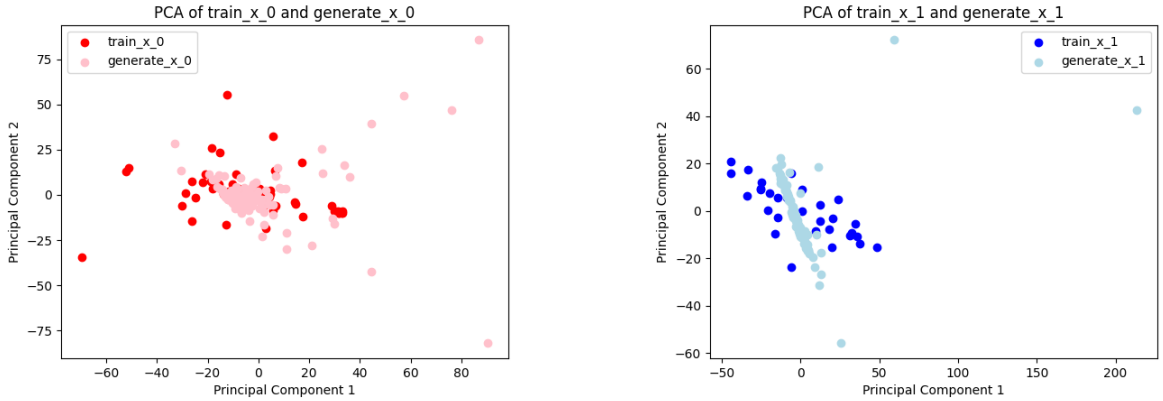


Fig. 16. Scatter plot of sample distribution for the USA dataset. (train_x_0 represents real unchanged samples, generate_x_0 represents generated unchanged samples; train_x_1 represents real changed samples, generate_x_1 represents generated changed samples). (a) Unchanged samples. (b) Changed samples.

TABLE IV
TEST TIME (S) AND PARAMETERS (M) ON EACH DATASET WITH ALL METHODS

Algorithm	parameters			TEST time		
	Farmland	River	USA	Farmland	River	USA
SVM	/	/	/	107.364	35.654	85.959
Hybrid	4.84	4.84	4.84	143.087	46.439	163.278
GETNET	9.37	15.42	9.22	171.105	264.775	180.297
CSANET	19.3	19.48	19.29	20.199	29.645	19.177
BiG-FSLF	2.38	2.38	2.38	18.836	26.207	18.067
FSCF-SSL	23.6	23.92	23.66	60.232	81.946	55.924
3DCNN-NF	0.13	0.17	0.13	272.149	494.504	317.731

E. Hyperspectral Tensor Sample Generation Experiment

In this article, we employed the normalizing flow technology to generate hyperspectral tensor samples to address the issue of a few training samples. To evaluate the effectiveness of this generative method, we compared it with other methods suitable for high-dimensional tensors, namely, rotate, Gaussian noise, and GAN. We utilized the basic CD model to generate CD results using four times the number of samples obtained through different generation methods. The CD results for samples generated by different methods on the three datasets are presented in Table V.

Figs. 8–10 illustrate examples of two generated difference samples along with their labels for each dataset. The experimental findings demonstrate that the sample generation method based on normalizing flow yielded the best CD results across all three datasets. This outcome highlights the significance of generating approximately identically distributed samples in facilitating the model's ability to learn category features.

To enhance the intuitive representation of the distribution, we utilize the PCA algorithm to extract the two principal components from both real and generated samples. We visually represent the sample distribution through scatter plots. Figs. 11–13 show the scatter plots of sample distribution for three datasets. It is evident that the distribution of the majority of generated samples aligns well with that of real samples. Nevertheless, owing to the approximate nature of the learned distribution, certain samples still exhibit notable deviations.

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TABLE V
CD RESULTS OF SAMPLES GENERATED BY DIFFERENT GENERATION METHODS

Algorithm	Index	Dataset Farmland	River	USA
Rotation	OA	0.9586	0.9321	0.9379
	Kappa	0.9088	0.6957	0.8215
	F1	0.9367	0.7014	0.8600
Gaussian Noise ($\mu = 0, \sigma = 1$)	OA	0.9638	0.9442	0.9514
	Kappa	0.9145	0.7150	0.8585
	F1	0.9404	0.7447	0.8896
Gaussian Noise ($\mu = 0, \sigma = 0.1$)	OA	0.9698	0.9409	0.9532
	Kappa	0.9280	0.7054	0.8611
	F1	0.9496	0.7366	0.8908
Gaussian Noise ($\mu = 0, \sigma = 0.01$)	OA	0.9716	0.9584	0.9531
	Kappa	0.9321	0.7678	0.8639
	F1	0.9523	0.7905	0.8940
GAN	OA	0.9717	0.9553	0.9458
	Kappa	0.9322	0.7562	0.8373
	F1	0.9522	0.7804	0.8714
Normalizing Flow	OA	**0.974**	**0.9691**	**0.9544**
	Kappa	**0.9366**	**0.801**	**0.8667**
	F1	**0.9549**	**0.8178**	**0.8958**

Consequently, in the labeling process, we utilize soft labels alongside label smoothing techniques to enhance the model's robustness against samples with larger deviations.

F. Parameter Sensitivity Analysis

This section conducts parameter sensitivity analysis experiments on the sample expansion number Q_k and the sample evaluation threshold θ involved in the algorithm.

1) Number of Sample Expansion Q_k: The number of sample expansions Q_k determines the proportion of real and generated samples in the final training samples. The larger Q_k is, the more the total number of training samples is, but the lower the proportion of real samples is. Therefore, we set $Q_k = \text{card}(\text{TR}_k) \times q, q \in N^*$, and evaluate the value of Q_k through the CD results on the three sets of datasets. Fig. 11 shows the experimental results of Q_k value on three datasets. The results show that when $q = 4$, the base CD model has the best performance on the three datasets.

2) Sample Evaluation Threshold θ_k and Label Smoothing Hyperparameter ϵ: The sample evaluation threshold, denoted as θ_k, is utilized to filter the generated samples that meet the criteria for the basic CD model. Larger values of θ_k generate samples that are more likely to be accepted by the base CD model, but they also make the generation process more challenging. The sample smoothing hyperparameter ϵ is employed to modify the soft labels, with a stronger adjustment associated with larger values of ϵ. Accordingly, we consider values within the range of $\theta_k \in [0.5, 0.9]$ and $\epsilon \in [0, 05]$ and assess the parameters based on the F_1 score of the base CD model. To assess the impact of soft labels, Fig. 14 presents experimental results for various parameter values on three datasets under the condition of hard label. The experimental results for the three datasets indicate that $\theta_k = 0.9$ performs better under hard label conditions. Fig. 15 presents the experimental results for three datasets, comparing different parameter values under the conditions of soft labels and label smoothing. The experimental results demonstrate a significant reduction in the value of θ_k when soft labels are introduced. This observation indicates the model's sensitivity to sample labels. Hard labels necessitate samples that exhibit prominent category features, whereas soft labels allow for labeling samples with weaker category features as well. Fig. 16 illustrates that when $\epsilon = 0.1$, the observed effect is more favorable. This suggests that the utilization of smooth labels enhances the model's generalization performance.

V. CONCLUSION

In this article, we present a few-shot HSI CD approach based on a 3D-CNN and normalizing flow 3DCNN-NF to address the challenge of limited training samples in the HSI CD task. We design a low-parameter CD network using 3D-CNN to mitigate the potential overfitting issue in the presence of limited training samples. In addition, we employ the normalizing flow technique to capture the spectral–spatial distribution of unchanged and changed hyperspectral tensors and generate augmented samples for training the CD network. Moreover, we lower the sample acceptance threshold by integrating soft labels and label smoothing methods, thereby enhancing sample generation speed while preserving the CD effectiveness. Experimental results demonstrate the superiority of our proposed algorithm over the majority of existing approaches in few-shot CD.

Nonetheless, this study is not without its limitations. Notably, the process of fitting the known label sample distribution through normalizing flow and filtering feasible samples is time-consuming. Moreover, the algorithm necessitates separate fitting for each class. Furthermore, the algorithm employed for the soft labeling of generated samples may lack the requisite accuracy, potentially impeding the training of the ultimate CD model. Consequently, future research avenues encompass the exploration of efficient methodologies for fitting high-dimensional tensor sample distributions and attaining more precise annotations for generated samples.

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